



Developing Partnerships and Negotiating Data Access

Session 3

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NOMMON



Overview

This session covers the following topics:

- **Incentives** and **disincentives** for Mobile Network Operators sharing MPD
- Different **examples of Data Access Models** and their main characteristics
- Different examples of **formalised agreements** for data access
- **Good practices** when accessing MPD

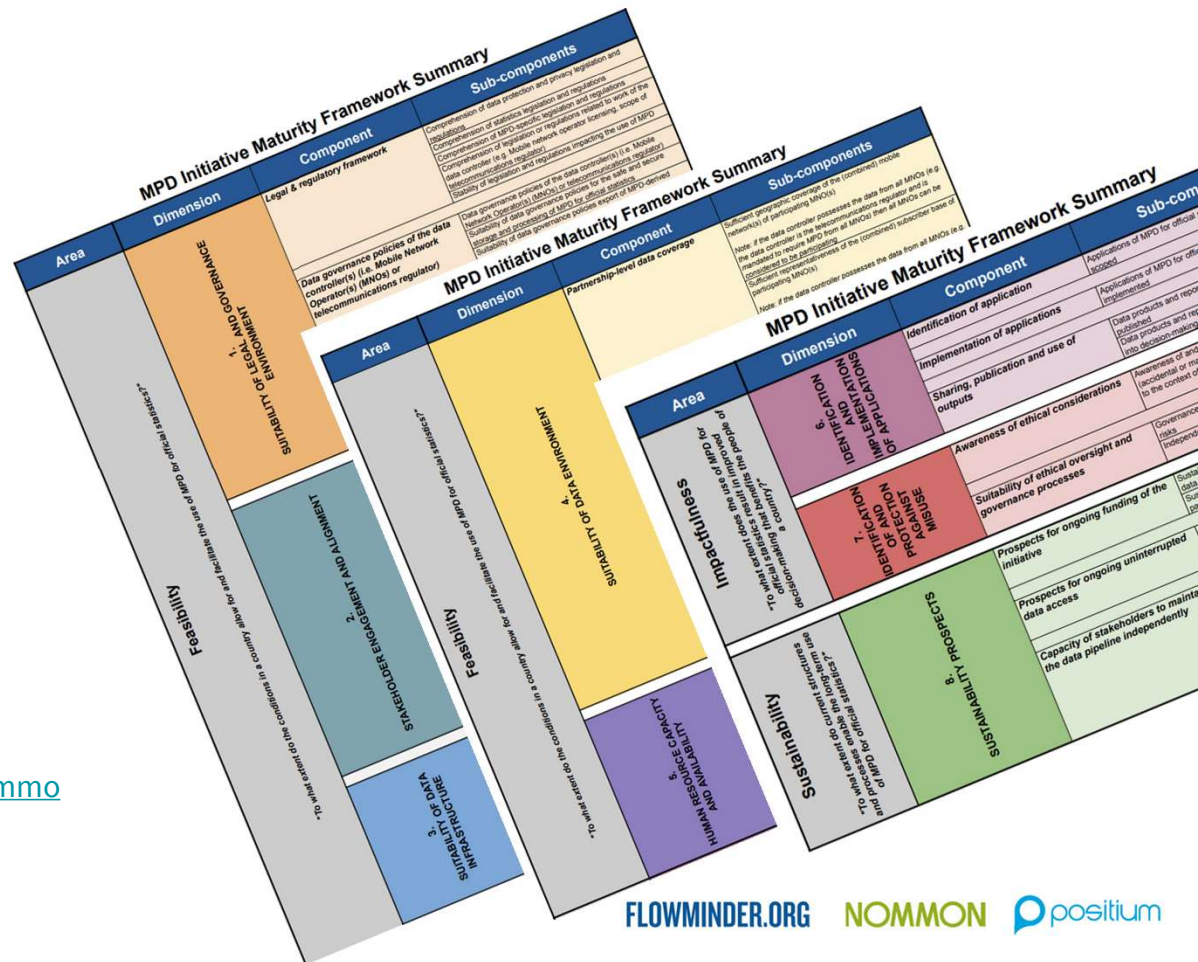
Resource

Maturity Assessment Framework & Tool

- Provides the tool for assessing readiness for mobile phone data projects along several dimensions.
- Helps analyse the legal situation and stakeholder capacity and support
- Identifies gaps that influence decisions regarding data access modes

Access the Maturity Assessment Framework and Tool here:

https://wbg.sabacloud.com/Saba/Web_spf/NA1PRD0002/commo n/resources/resourcedetail/simrs000000000015427



Resource

A Roadmap to Accessing Mobile Data for Statistics

- Provides a step-by-step guide for NSOs dealing with mobile network data access to improve official statistics.
- Uniquely, this guide prioritizes **addressing the non-technical challenges** that often stall data sharing projects, such as effective communication and stakeholder alignment.
- Includes **successful examples** from countries that have navigated the path to accessing mobile network data, offering insights and best practices.



Access the manual on accessing Mobile Phone Data from here: <https://www.data4sdgs.org/roadmap-accessing-mobile-network-data-statistics> (Note: The content can be downloaded as a PDF from this site, [here](#). Also available in [French](#) and [Spanish](#))

Negotiating data access with Mobile Network Operators (MNOs)

Role of Mobile Network Operator

- Data holder:
 - Original data source, can provide access to data
- Can leverage:
 - internal knowledge on the data source
 - technical capacity (IT)
- For ongoing, sustainable data pipelines, needs to provide:
 - Regular access to new CDR data, including all associated data fields
 - Regular updates to cell tower location data
 - Fixes when pipeline is disrupted (e.g. power outages, data gaps)
- Potential gains for MNO:
 - Benefit from statistical expertise of NSO
 - Gain access to other data sources at the NSO
 - Brand, reputation and corporate social responsibility
 - Financial incentives



Potential Disincentives for MNOs

Compliance issues

1. Legal risk – Risk of not complying with all regulations, especially telecommunications and data protection
2. Public opinion risk – Civil society or the public speaking up against use of subscriber data

Capacity issues

1. Low technical readiness
2. Low human capacity

Business issues

1. Investments involved – Changes in data collection, establishing servers and staff to enable data flow, etc.
2. Competing with own data monetization efforts – Why give data for free if we can sell it

Compliance can be achieved through formalized agreements and data governance

Capacity can be raised or provided

Business decisions require a balance between costs and benefits

Potential Incentives for MNOs

Indirect Benefits from Collaboration in Statistics

Regulatory compliance & legal mandates



Statistical or telecommunications law making it mandatory for MNOs to share data.

E.g. Having a clear legal framework for sharing data to the government, compels the MNO and reduces their data sharing risk.

Public Recognition & CSR



Acknowledgment of MNOs' contribution to creating statistics as a public good; boosting their corporate social responsibility profile.

E.g. MNO has a publicly stated goal to contribute to the development of the society.

Data Reciprocity



Receive data from statistical offices, compare/merge MPD with other data (surveys, admin data) and/or receive new insights.

E.g. MNO can use fine-grained demographic data from NSOs to improve marketing, business strategy, and distributing funds for the development of infrastructure.

Potential Incentives for MNOs

Direct Benefits to Increased Capacity

New Products & Monetization



Explore potential revenue streams from data-driven products and services beyond those of official statistics.

E.g. MNO uses the same high quality pipeline used in statistics for commercial data provision.

Data Analysis Capacity Building



Build internal expertise and infrastructure for data analysis; Access NSO and external expertise for capacity building.

E.g. MNO technical staff take part in global or national training on MPD analysis methods and algorithms.

Innovation and R&D

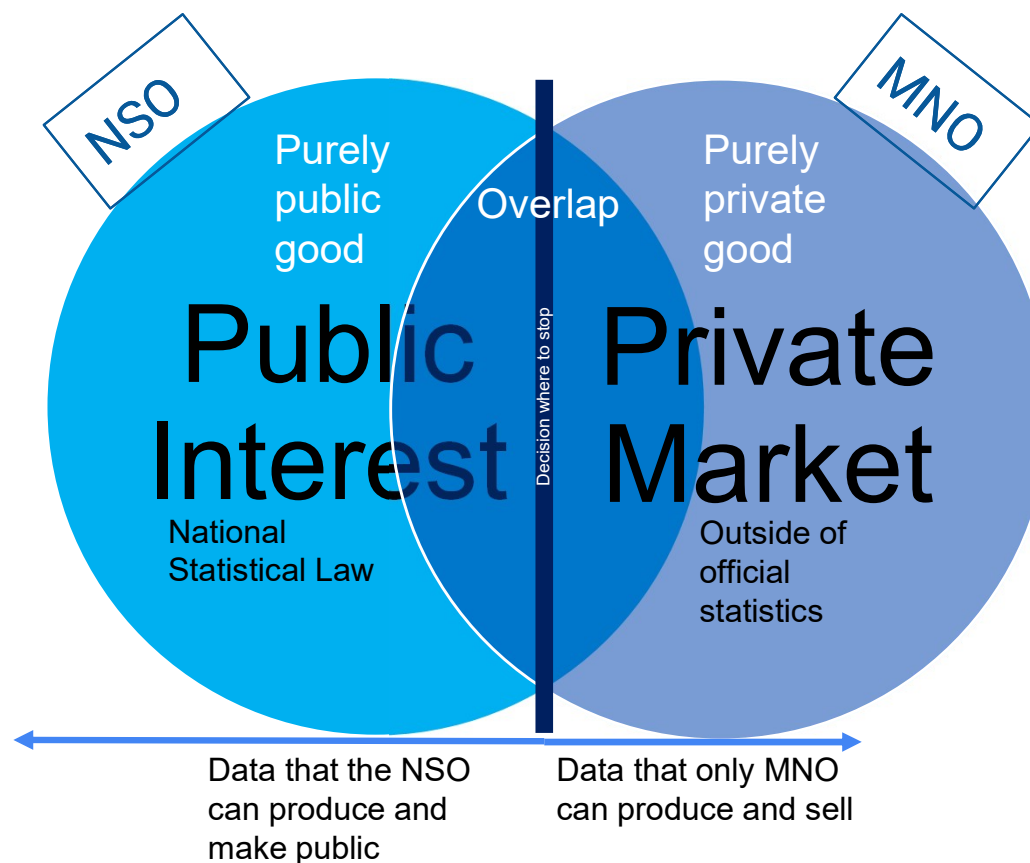


Improvements to MNO data products and/or services. Increase in quality, representativeness and/or robustness of outputs.

E.g. MNO will improve how much data they collect, how they process it and how they assess the quality, through investments in the project.

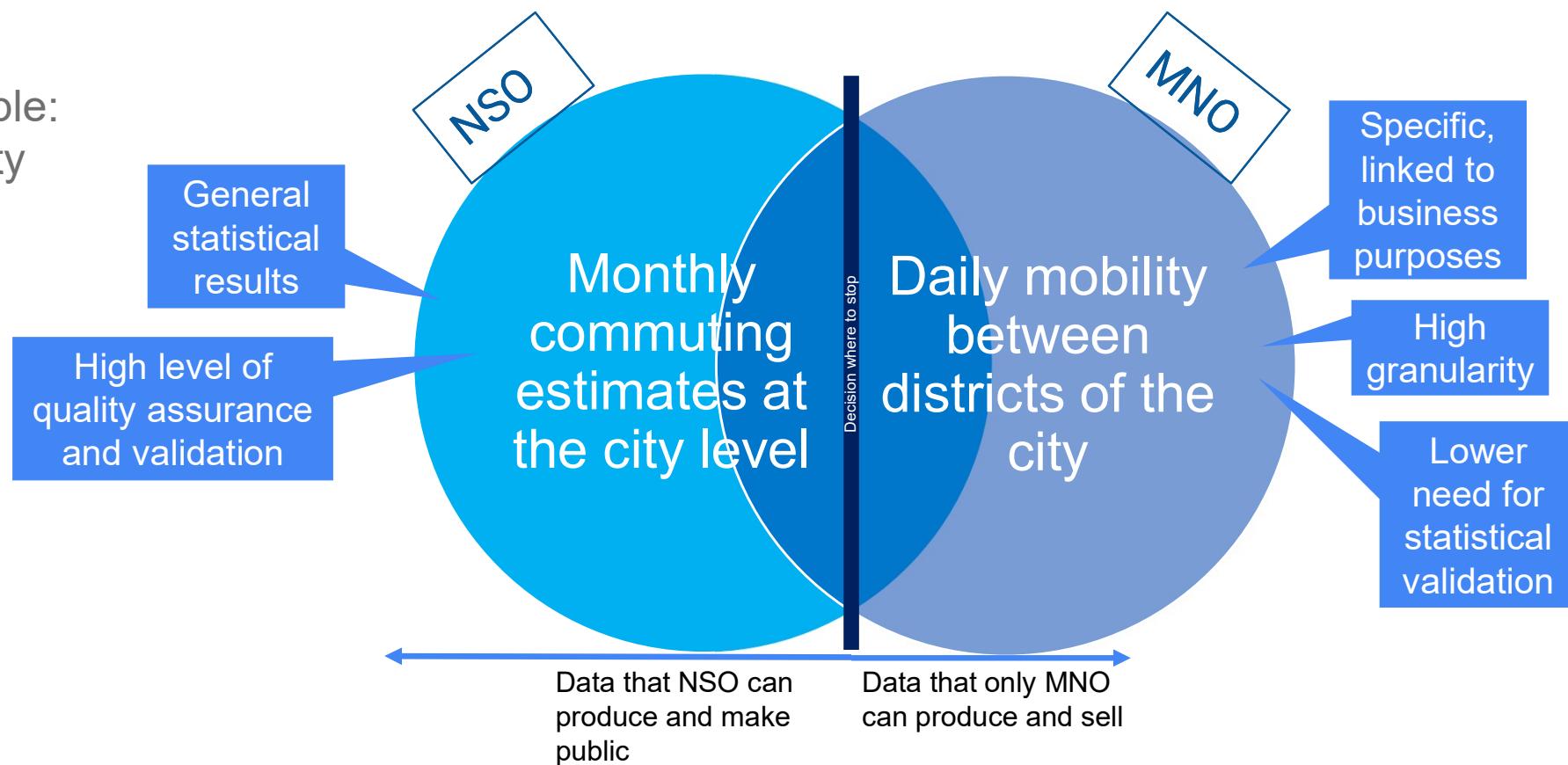
Agree What Data is in Public Interest

Agree on the division between markets –
How much is in the public interest and how much data analysis should be left for the MNO to monetize?



Agree What Data is in Public Interest

Example:
Mobility



Consolidating Activity

Reflection: Data Sharing Incentives and Disincentives

Think about a real or hypothetical MNO in your country.

1. What are its potential incentives to participate in this project (see below)?
2. What are the likely disincentives or risks for this MNO that you, as the NSO, would need to address or mitigate (e.g., data privacy concerns, financial costs, resource strain)?

Regulatory Compliance



Public Recognition



Enhanced Data Analysis Capabilities



Data Products, Monetization



Data Reciprocity

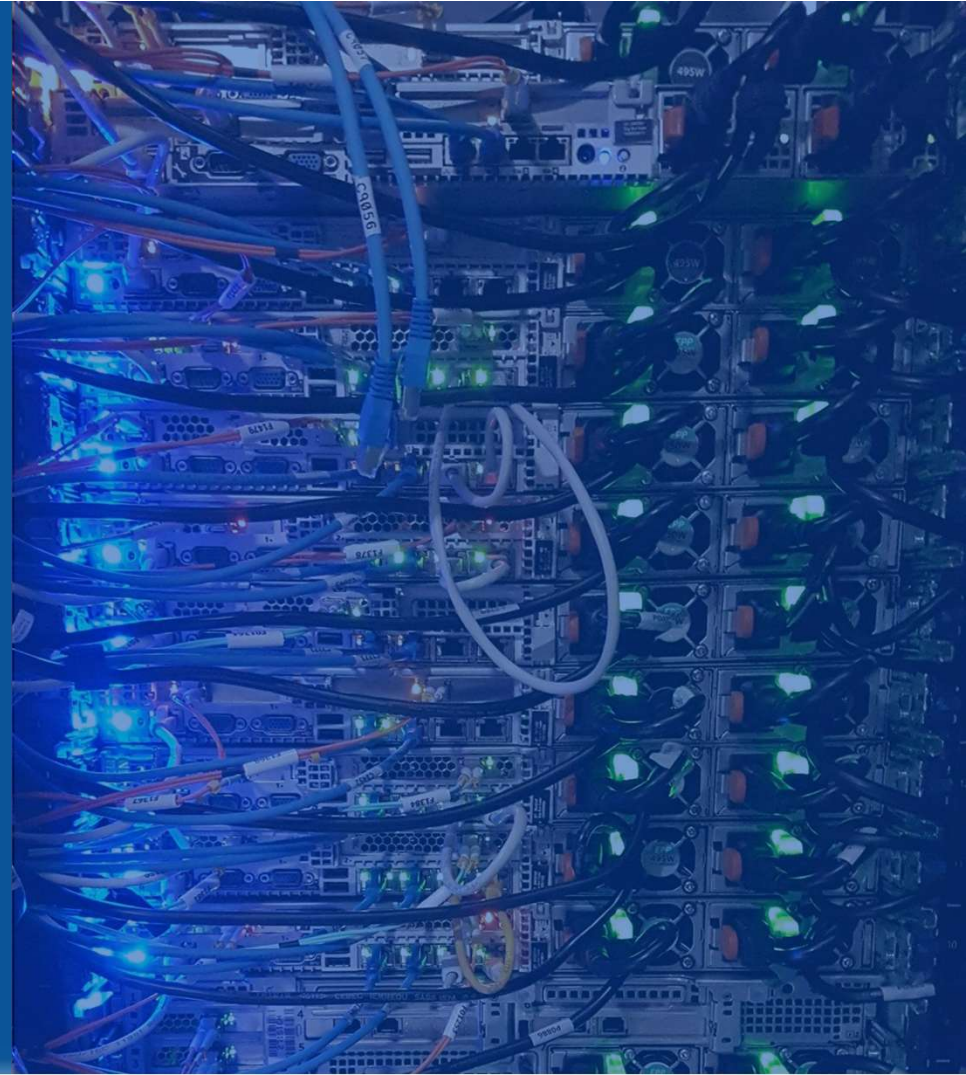


Innovation and R&D

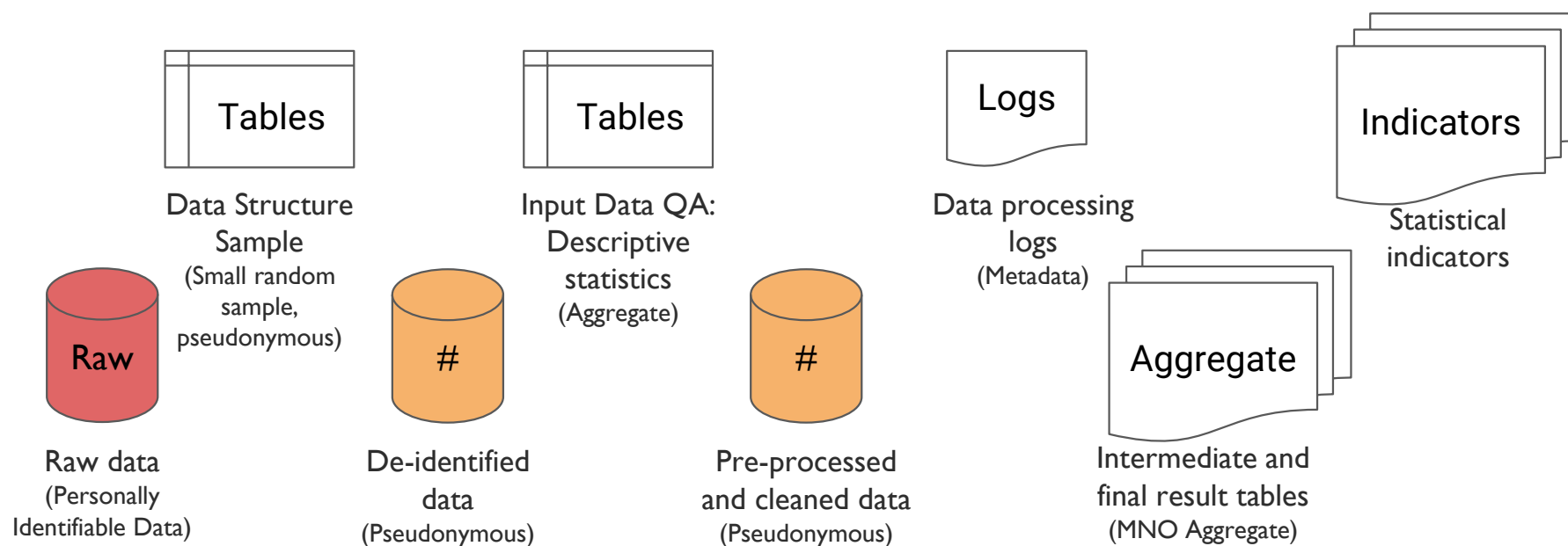


Data access models

3 best practice data access scenarios



Types of Datasets in an MPD Project



Types of Technical Roles

Statistical Business Process Terms

It should be clearly identified which entities:

- **Collect data.** For MPD, this is usually the MNO.
- **Design and build** the data quality framework and processing algorithms. For MPD, this is usually the NSO together with a technical service provider.
- **Process** the data. For MPD, this is usually the MNO, a technical service provider, the NSO or Regulator.
- **Disseminate** the data. For MPD, this is usually the NSO or the user Ministry or Department.

Collect	Process	Design, Build, Analyze	Disseminate, Evaluate
Who?	Who?	Who?	Who?
What?	What?	What?	What?

Types of Legal Roles (in Data Protection)

General Data Protection Terms

The **data controller** determines the purposes for which and the means by which personal data is processed.

The **data processor** processes personal data only on behalf of the controller.

The duties of the processor towards the controller must be specified in a **contract or another legal act**.

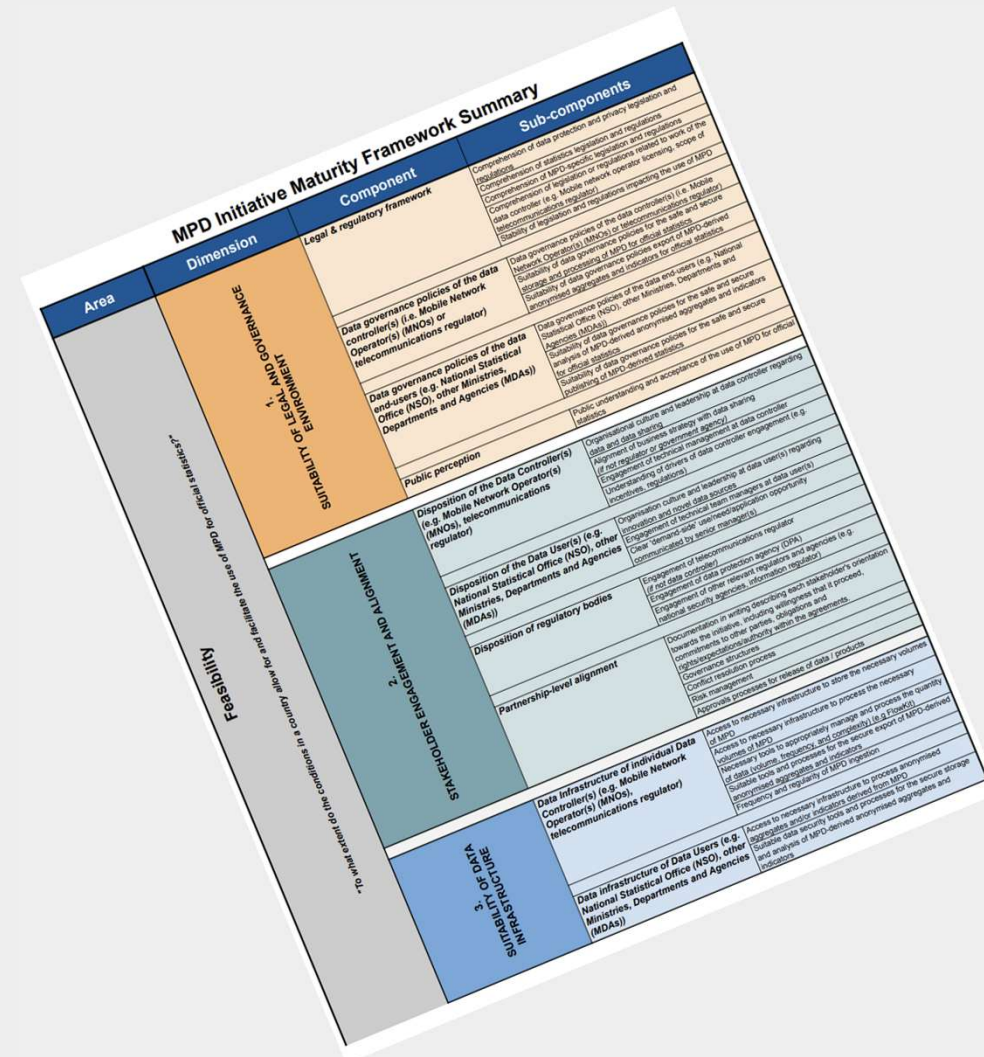
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Assessing Maturity

Suitability of the legal environment, stakeholder and infrastructure capacity, and security



Legal Feasibility: Mandate and Authority

The first and most critical criterion is the legal framework. This question determines whether the NSO has the legal authority to even act as a data controller or processor.

Question: Does the NSO's legal mandate or a specific law authorize it to be a **Data Controller** for sensitive private-sector data, and can it compel MNOs to transfer this data?

- **If Yes:** The NSO can potentially collect and process the raw data itself.
- **If No:** The NSO is likely limited to a lower level of access, such as being a **Data Processor** or a data user. This moves the decision criteria to the next level.

Technical Feasibility: Capacity and Infrastructure

This criterion assesses the NSO's ability to handle the massive volume and complexity of MPD, including data security.

Question: Does the NSO possess the necessary **technical infrastructure and human expertise** to securely ingest, store, process, and analyze petabytes of raw, granular MPD from multiple MNOs?

- **If Yes:** The NSO has the capacity to be a processor of the raw data. This supports a higher level of access.
- **If No:** The NSO may need to rely on the MNO or a technical service provider for data processing. This restricts the NSO's role to designing the processing tools, validating the output, or simply using the final statistical product. This moves the decision criteria to the next level.

Organizational Feasibility: Governance and Public Perception

This criterion focuses on the NSO's ability to maintain public trust and ensure data confidentiality, which is paramount for sensitive MPD. The most vague criterion, and usually trust is established through reputation, collaborative history, support from high-level authorities overseeing communications and data protection, information security standards and well-structured contracts.

Question: Can the NSO establish and **maintain a level of public and political trust** that assures the MNOs and the public that the sensitive individual-level data will be handled with the highest standards of **confidentiality and security**?

- **If Yes:** The NSO is considered a trustworthy steward of the data, which facilitates greater access to processed or even raw data.
- **If No:** The NSO's access is more likely to be limited to aggregated, anonymized, and pre-processed data to minimize privacy risks. This moves the decision criteria to the next level.

Levels of Access for NSO (Government)

Data User

Level 1: NSO as a Data User and Disseminator only. The NSO **lacks the legal mandate or technical capacity** for raw data processing. The MNOs or a service provider process the data according to the NSO's quality standards. The NSO's role is to **validate the final, aggregated outputs** and disseminate the statistics. This is the most common model when legal or technical hurdles are significant.

Example: Ghana

Data Processor

Level 2: NSO as a Processor and Designer. The NSO may not be the data controller but **has the technical capacity and trust** to access a pre-processed data stream from the MNOs or a technical service provider. The NSO can **design the processing algorithms and quality framework** and conduct final statistical validation before dissemination.

Example: Indonesia

Data Controller

Level 3: NSO as Data Controller and Processor. The NSO has the **legal mandate, technical capacity, and public trust** to receive raw, granular MPD directly from MNOs, process it into statistical outputs, and then disseminate it. This is the highest level of access.

Example: Oman, Estonia

MNO Hosting, Authorized Data Processing

MNO creates a sandbox for NSO to develop methodologies on de-identified raw sample data. Methodology and data processing overseen by NSO, potentially assisted by external parties. Data processing transferred fully to MNO for production.

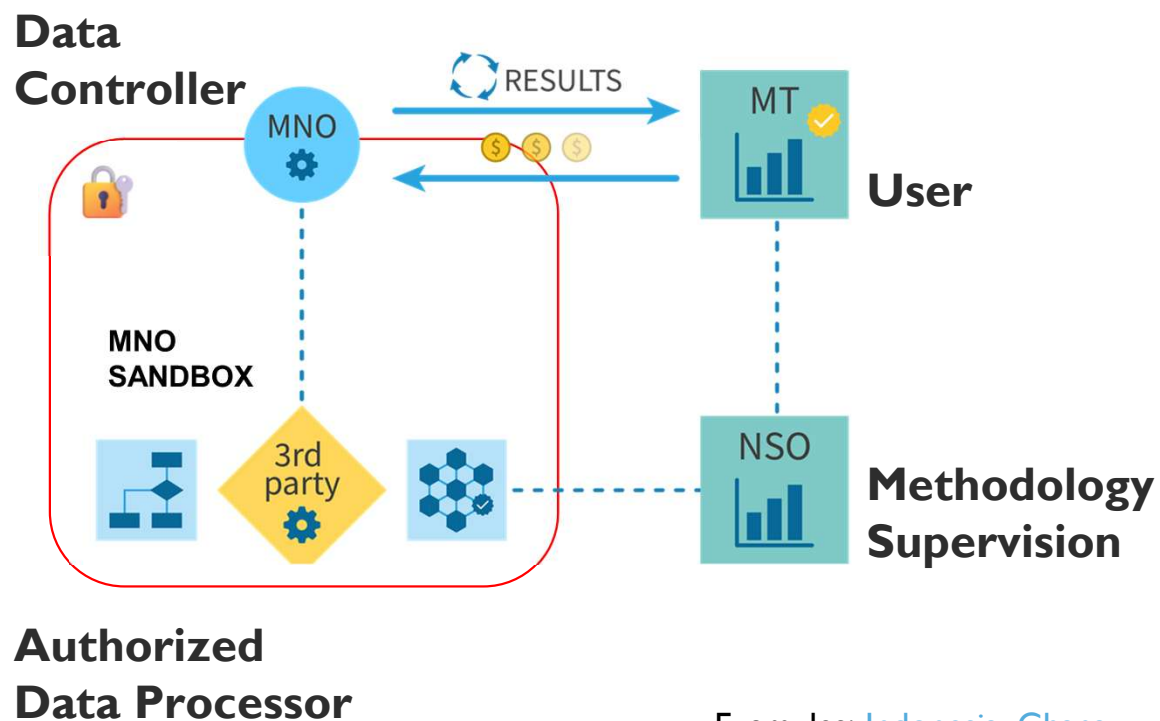
MNO = Data Owner & **Data Controller**

NSO = Supervisory Authority for Statistical Quality and Methodology, Data User and/or processor of sample data

Processing party = **Authorized Data Processor for Implementation**, until handover

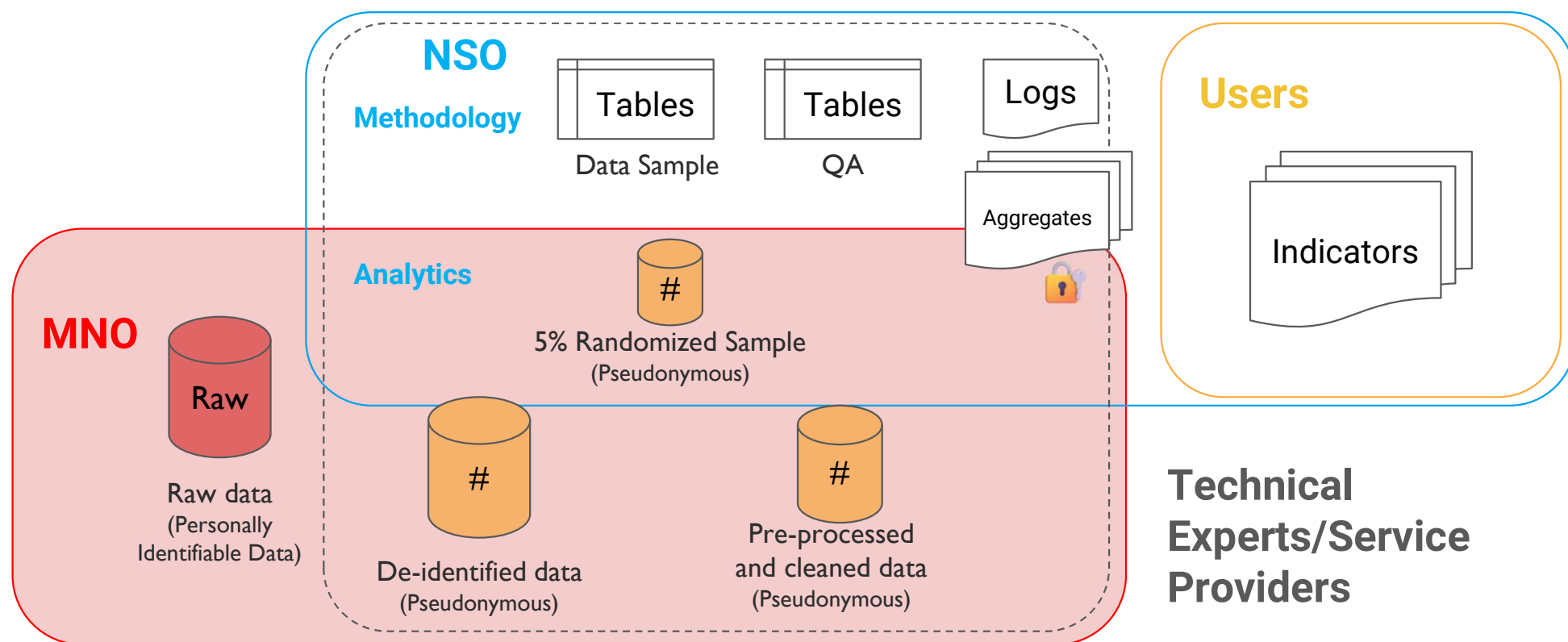
MNO = **Data Processor during production**

Ministry = Data User (and potential financier)



Examples: [Indonesia](#), [Ghana](#)

MNO Hosting, Authorized Data Processing

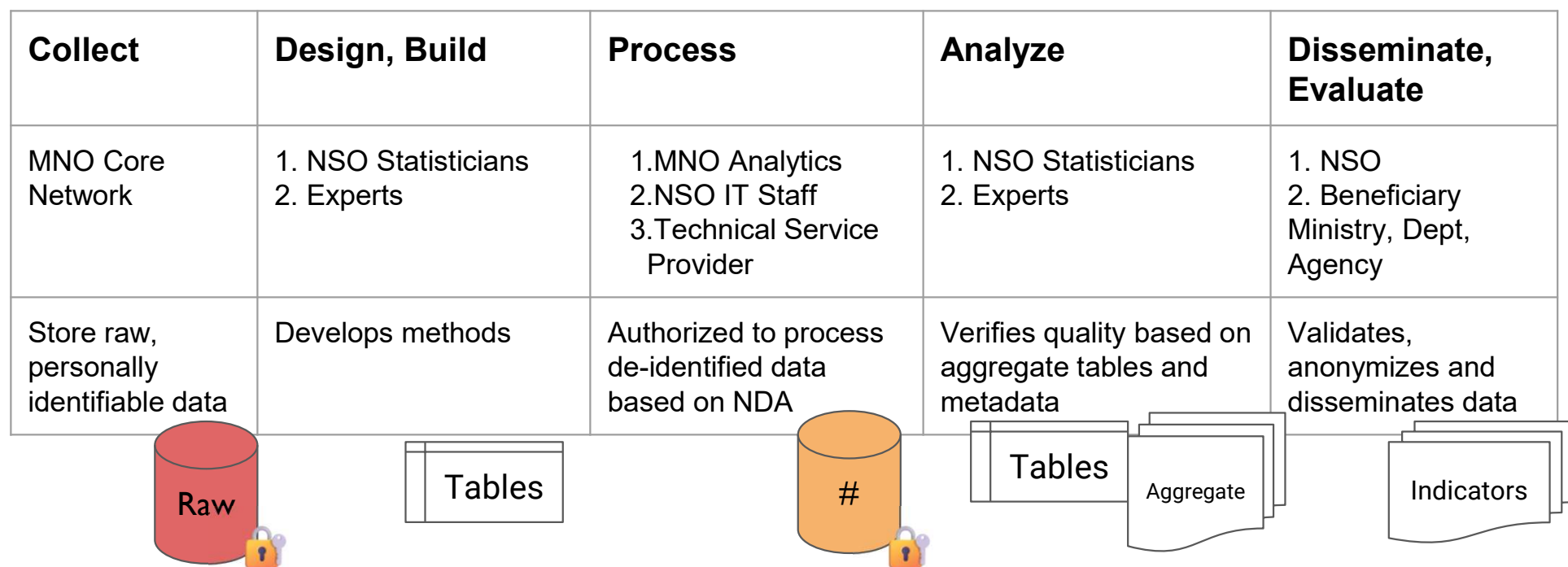


MNO Hosting, Authorized Data Processing

Distribution of responsibilities in the cross-border tourism statistics project in Indonesia, 2017-2022.

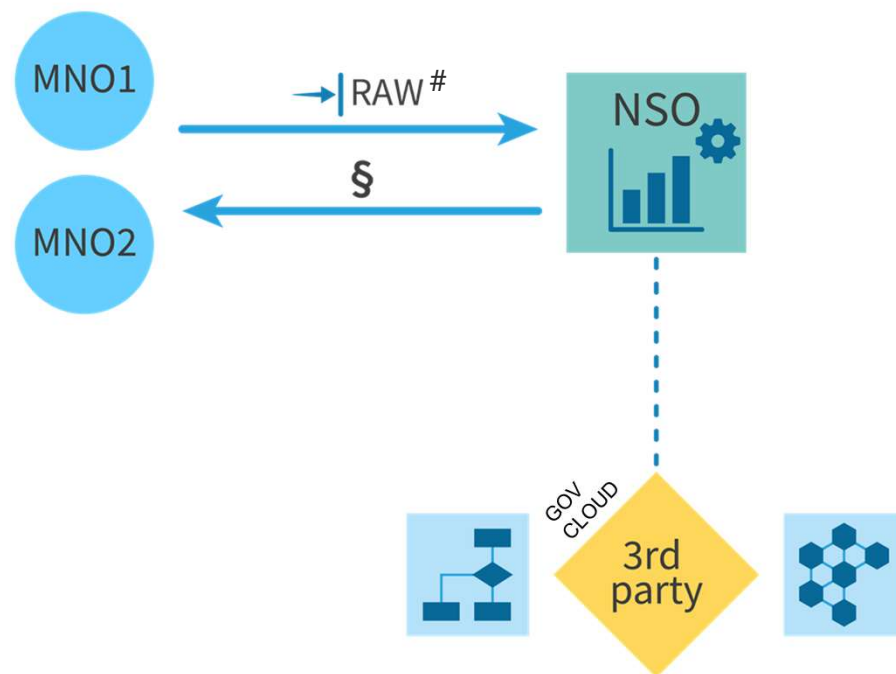
Business Process	NSO	Technical Service Provider	Mobile Network Operator
Specify needs			
Design			
Build			
Collect			
Process			
Analyze			
Disseminate			
Evaluate			

MNO Hosting, Authorized Data Processing



Examples: [Indonesia](#), [Ghana](#)

NSO as the Data Controller



De-identified raw data transfer from MNOs to a central location, for example a secure Government Cloud. Data processing overseen by NSO, potentially assisted by external parties.

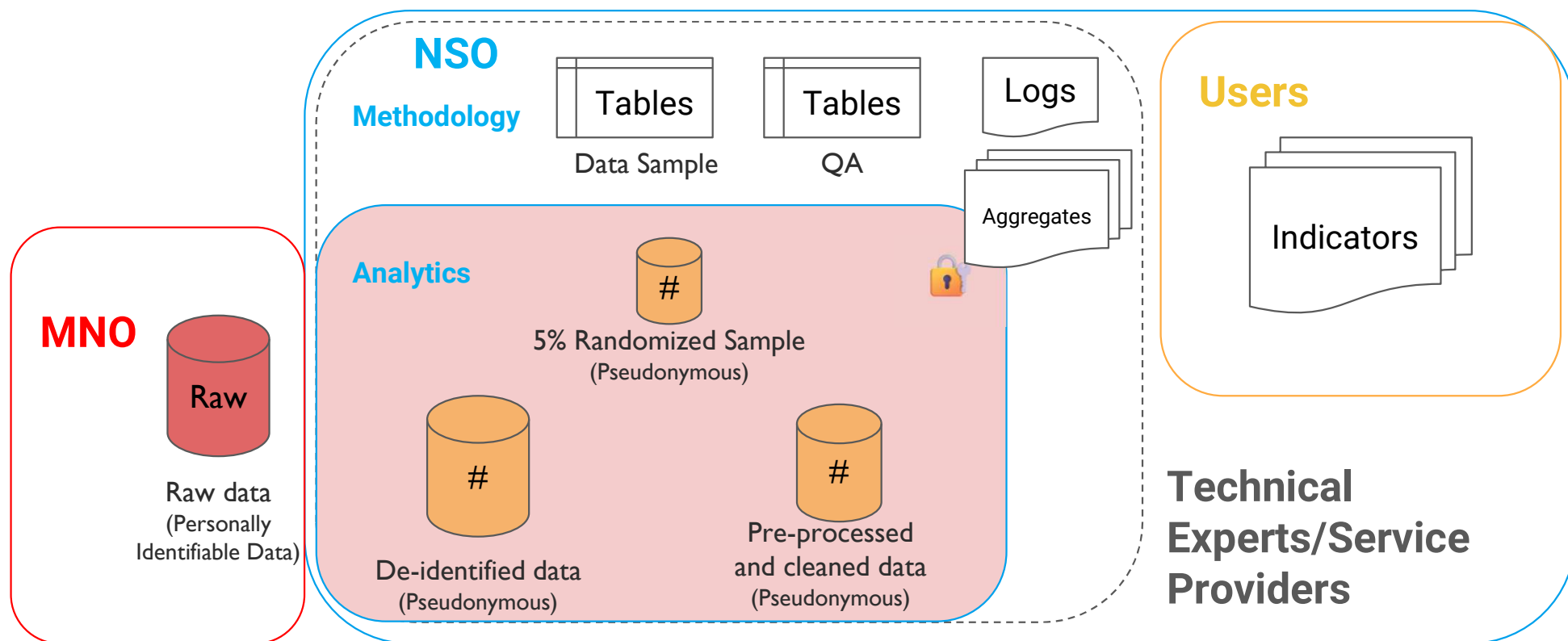
MNO = Data Owner

NSO = **Data Controller**

Processing party = **Authorized Data Processor**

Examples: [Oman](#), [Estonia](#)

NSO as the Data Controller



Regulator as the Data Controller

IF the telecom regulatory authority has the regulatory mandate and technical capacity. NSO oversees methodology development and implementation, potentially assisted by external parties.

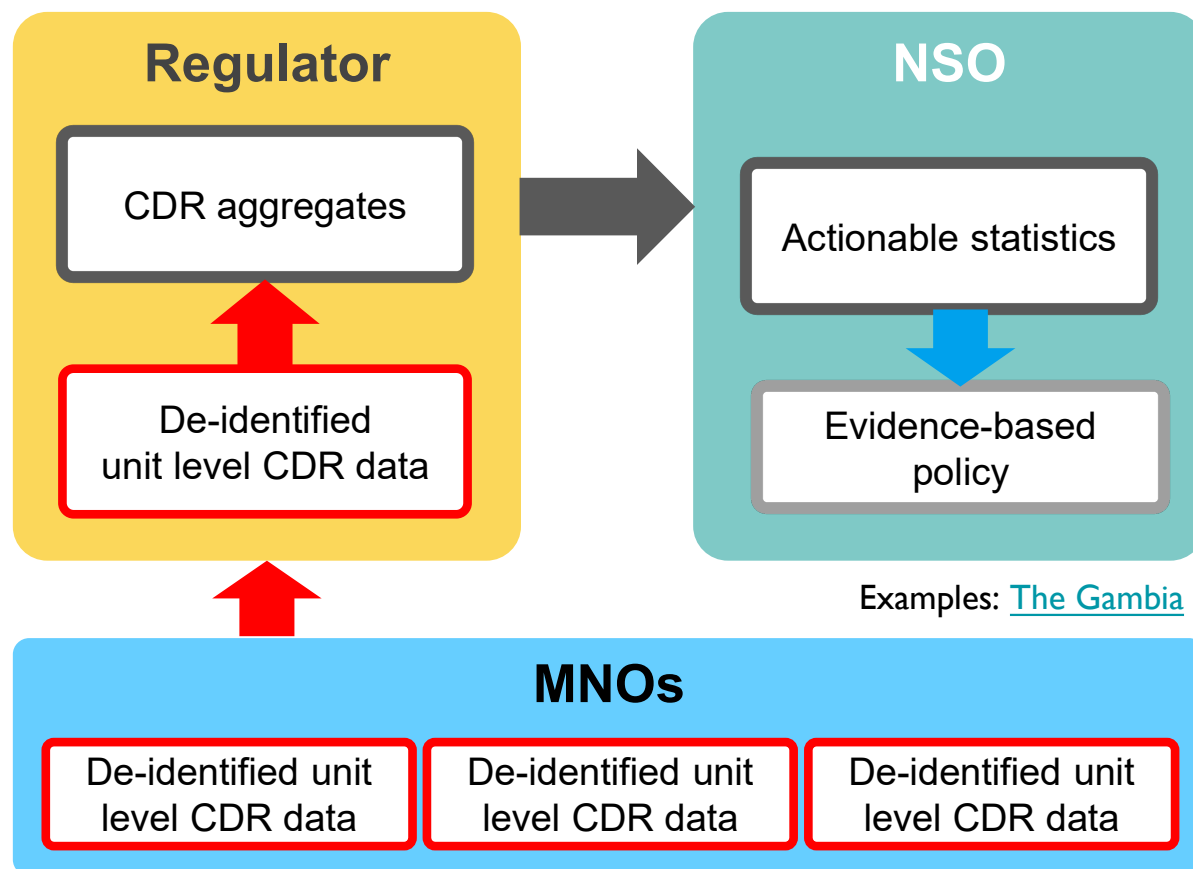
MNOs = Data owners

Other party = **Data controller**

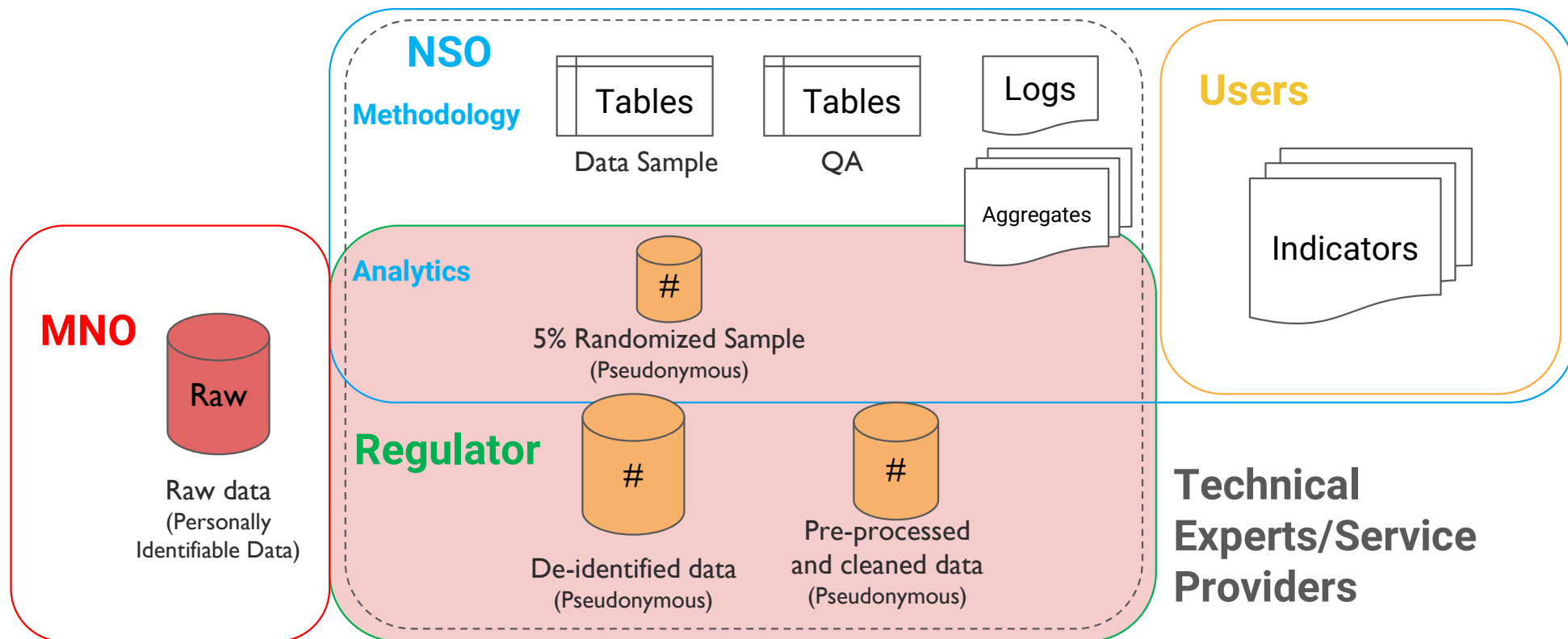
Technical Service Provider =

Authorized Data Processor

NSO = Data User



Regulator as the Data Controller



Reflection on Data Acquisition

Considering the real-world context of your country (e.g., legal framework, data privacy laws, MNO technical maturity, NSO capacity, potential technical service providers), which data access model would be the most suitable?

1. Who will be the data controller?
2. Who will receive or be able to access the data, being authorized to process data?
3. Who handles methodology development and auditing function?

Justify your choice by explaining what main gaps in capabilities exist and how it fits the situation you've described in the previous question.

Agreements for data sharing

Legal Basis for Data Access -
Setting up MOUs and Data Sharing
Agreements

AGREEMENT

BETWEEN

[MNO]

AND

[NSO]

AND

*[OTHER PARTNER/S (E.G TECHNICAL SERVICE
PROVIDER)]*

Types of Data-Sharing Documents for MPD Access

Non-Disclosure Agreement (NDA)

Focused on **confidentiality**.

Can be used to allow access to data, as long as the obligations, scope and terms of data processing are included.

Can be signed by individuals for personal access.

Memorandum of Understanding (MoU)

Focused on **collaboration principles**.

The MoU might describe the general goals and principles of the data sharing.

Can be signed for data access in case there is a legal basis.

Data-Sharing Agreement (DSA)

Focused on **data sharing rules**.

A legally binding document that includes all rules and protocols that govern sharing of data

Can be used in case a comprehensive legally binding document is needed.

Legal Mandate > MoU, Sultanate of Oman

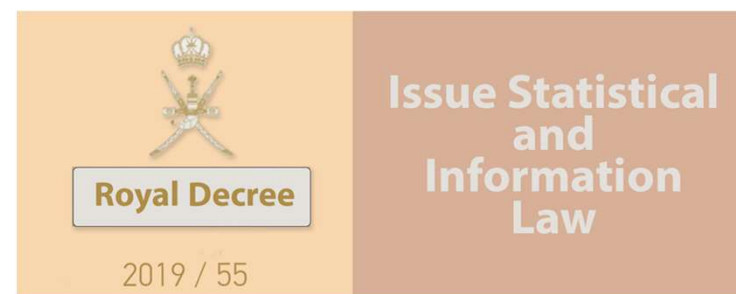
A Royal Decree issued in 2014 to regulate the work of the Centre

- Granting broad powers to the Centre in collecting and storing official statistics produced by the Centre and other government and non-government agencies



A Royal Decree issued in 2019 regarding the Statistics and Information Law:

- Article 19 In order to achieve its objectives, the Center may request whatever data, information and administrative records it deems necessary from individuals, governmental and non-governmental agencies, without fees



Article 32 punishes every non-governmental agency that refuses to provide the data and information required in accordance with the provisions of this law that are required by those in charge of official statistical activities

Memoranda of understanding with telecom companies

MoU > Contract > NDAs, BPS Statistics Indonesia

No clear legal mandate in Statistical Law.

MoU between BPS and Telkom was signed in 2018 - establishes cooperation principles.

Commercial contract between BPS and Telkom was signed in 2018 - establishes requirements for sample data access in an MNO-hosted environment.

Personal NDAs signed by BPS data scientists to access data in MNO environment.



NOTA KESEPAHAMAN

ANTARA

BADAN PUSAT STATISTIK

DAN

PERUSAHAAN PERSEROAN (PERSERO)
PT TELEKOMUNIKASI INDONESIA Tbk

TENTANG

**PENYEDIAAN, PEMANFAATAN, DAN PENGEMBANGAN
DATA/INFORMASI, TEKNOLOGI INFORMASI, DAN KOMUNIKASI**

Berdasarkan pertimbangan sebagaimana tersebut di atas, **PARA PIHAK** sepakat untuk mengadakan Nota Kesepahaman tentang Penyediaan, Pemanfaatan, dan Pengembangan Data/Informasi, Teknologi Informasi, dan Komunikasi, yang selanjutnya disebut Nota Kesepahaman, dengan ketentuan sebagai berikut:

Pasal 2 RUANG LINGKUP

Ruang lingkup Nota Kesepahaman ini meliputi:

- Penyediaan dan pemanfaatan Internet dan *Managed Service* (VPN IP, Astinet, Wifi, Mobile CUG, *Managed Network Service*, *Machine-to-Machine/ M2M*);
- Pengembangan *Data Center* atau *Data Recovery Center (DRC)*;
- Penyediaan, pemanfaatan, dan pengembangan data/informasi dalam bidang teknologi informasi dan komunikasi;
- Pengembangan sumber daya manusia di bidang statistik, teknologi, informasi dan komunikasi; dan
- Kerja sama lainnya yang disepakati oleh **PARA PIHAK**.

Data-Sharing Agreement, Ghana



USING MOBILE DATA FOR HEALTH MONITORING:

A Case Study of Data Sharing Between Ghana Statistical Services, Vodafone Ghana, and Flowminder Foundation

A Case Study by SDSN TReNDS for C4DC



Important agreement terms

Data Anonymization and Aggregation - the agreement specifically includes precautions on how data should be aggregated and which categories of data the GSS can access

Data Handling – responsibilities of the different parties for analyzing data

Data Use Limitations – Use Vodafone Ghana's data only for the stated purposes

Data Deletion – required to return or delete

Publishing Results – release outputs under an open source license

GDPR Requirements – establishing procedures under Data Processing Agreements



Consolidating Activity

Reflection: Agreements for Data Sharing

Data sharing agreements

For the data sharing initiative in your country context, which mechanisms are already in place, or need to be described in a written agreement, to enable data to be accessed?

- Non-disclosure agreements (NDAs);
- MOUs or collaboration agreements; and/or
- Data-sharing agreements (DSAs).

See also session: Data Governance

Every MPD initiative will be different

There will always be aspects that differ depending on the local context

Summary

1. It is important to **understand what incentives and disincentives Mobile Network Operators have to share data**. If clearly acknowledged, they can be strengthened or mitigated.
2. Various models have been successfully used to produce statistics with mobile network data, ranging from full to no raw data transfer. **The maturity of partners in the project determines, which data access model is preferred.**
3. Partnerships for data access are formalized in agreements, either through:
 - a. Non-disclosure agreements (NDAs) to ensure confidentiality, even on the individual level.
 - b. Memorandum of Understanding (MOU) to set up a general working relationship or describe data sharing in case of legal basis, and/or
 - c. Data-sharing agreements (DSAs), which define all terms and conditions under which data sharing happens for a specific project.

Thank you!



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Mobile Phone Data for Policy Programme (WB-GDF)
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